

A unified empirical treatment of Deutsch closed-timelike-curve channels: amplification, cycles, multi-basin dynamics, and the encoding-dependence of chronology protection

Christian Knopp
cknopp@gmail.com

2026-05-11

Abstract

We present a sustained empirical investigation of Deutsch’s closed-timelike-curve formalism [1] on small Hilbert spaces, organised across 33 numerical phases covering roughly 10^5 sampled channels. The investigation yields six empirical regularities that collectively revise the textbook picture of D-CTC iteration: (1) The iteration count to convergence is log-normally distributed and predicted by $|\lambda_2(\mathbf{E})|$ at Pearson correlation $r = 0.99$, giving an $\mathcal{O}(d^6)$ spectral oracle for convergence prediction. (2) The channel class that produces Aaronson–Watrous-style [2] state-distinguisher amplification is not generic Haar randomness but a Clifford-like subgroup (permutation $\times$ diagonal of fourth roots of unity), at rate $\sim 100\times$ above Haar. (3) The empirical amplification reaches $13\times$ speedup over the Helstrom bound at close inputs ($\epsilon = 0.05$) and is robust to $\sim 10\%$ depolarising noise. (4) The majority of Clifford-class channels do not converge: they reach limit cycles of period k with $\arg \lambda_2(\mathbf{E}) = 2\pi/k$. Cycle solutions amplify $\sim 2\times$ more than fixed-point solutions. (5) D-CTC dynamics are multi-basin: a Clifford-class channel admits on average ~ 25 distinct limits depending on initial state, overturning the textbook ergodicity claim. (6) The chronology-protection conjecture’s signature in D-CTC amplification depends on the channel’s encoding: a smooth continuum from abstract (Clifford/Floquet) to direct-CTC (matrix entries proportional to the spacetime CTC tensor), with the chronology-protection signature emerging gradually above 50% direct-CTC content. Together these findings imply that Deutsch’s self-consistency equation should be re-read as admitting cycle solutions of any period $k \geq 1$, with the Aaronson–Watrous polynomial-time PSPACE speedup operating on cycle solutions through marginal-mode ($|\lambda_2| = 1$) information preservation. All code, data, and 33 reproduction scripts are at github.com/Zynerji/systrophe (v0.17.0). This treatise supersedes the two prior reports [6, 7] and serves as the unified record.

Contents

1	Introduction and motivation	1
2	Setup and notation	2
2.1	D-CTC iteration and channel superoperator	2
2.2	State-distinguisher amplification	2
3	Spectral oracle for convergence	2
3.1	Pearson correlation $r = 0.99$	2
3.2	Log-normal iteration distribution	3
3.3	Power-law dimensional scaling	3

4	Aaronson–Watrous amplification: empirical	3
4.1	Channel-class enhancement	3
4.2	Amplification scales with input closeness	3
4.3	Noise robustness	3
5	Limit cycle structure	3
5.1	Cycle-length distribution	4
5.2	Spectral prediction of cycle length	4
5.3	Cycles amplify more than fixed points	4
6	Multi-basin dynamics	4
6.1	Quantitative overturning of ergodicity	4
6.2	Adversarial probing	5
7	Chronology-protection coupling	5
7.1	Setup	5
7.2	Three encoding regimes	5
7.3	The encoding is a continuum	5
7.4	Implication	6
8	Mechanism: the Clifford structural signature	6
8.1	Bimodal eigenvalue distribution	6
8.2	Structurally degenerate U-blocks	6
8.3	Full Bell-state disentanglement	6
8.4	Higher Holevo classical capacity	6
8.5	Code-state-quality fixed points	7
9	Re-interpretation of the Deutsch equation	7
9.1	Properties of the cycle-solution class	7
9.2	Aaronson–Watrous on the cycle class	7
9.3	Numerical practice	7
10	Implications	7
10.1	Computational complexity	7
10.2	Physics interpretation	8
10.3	Numerical practice	8
11	Open questions	8
12	Reproduction	8

1 Introduction and motivation

Deutsch’s 1991 formulation of closed-timelike-curve quantum mechanics [1] converts the apparent paradox of CTCs into a fixed-point equation on density matrices. For a CPTP channel $\mathcal{E}$ acting on a joint Hilbert space $\mathcal{H}_{\text{CR}} \otimes \mathcal{H}_{\text{CTC}}$ (where “CR” denotes the chronology-respecting register and “CTC” the register that traverses the closed timelike curve), the self-consistency condition is

$$\rho^* = \mathcal{E}(\rho^*), \quad \mathcal{E} : \rho \mapsto \text{Tr}_{\text{CR}}[U(\sigma_{\text{CR}} \otimes \rho)U^\dagger], \quad (1)$$

where U is a unitary on the joint space and σ_{CR} is the CR input state. Aaronson and Watrous [2] showed that an oracle for ρ^* enables polynomial-time computation of PSPACE-complete

problems, in particular distinguishing arbitrarily close mixed states beyond the limits of the Helstrom bound [4].

In the standard treatment, (1) is read as defining a unique fixed point reached by iteration $\rho_{n+1} = \mathcal{E}(\rho_n)$. Empirically, this picture is incomplete in several important ways. Below we report 33 phases of numerical experiments that establish six empirical regularities. The combined picture is substantially different from the textbook reading of (1).

Throughout we work at $d_{\text{CR}} = 2$, $d_{\text{CTC}} = 3$ unless otherwise noted, giving joint dimension $d = 6$. Sample sizes are at least 10^3 per ensemble. The Systrophe package [8] provides the computational infrastructure and reproduction scripts.

2 Setup and notation

2.1 D-CTC iteration and channel superoperator

The iteration

$$\rho_{n+1} = \mathcal{E}(\rho_n) \quad (2)$$

is conducted with tolerance $\|\rho_{n+1} - \rho_n\|_F < 10^{-10}$, starting from a random initial pure state $\rho_0 = |\psi_0\rangle\langle\psi_0|$. The channel $\mathcal{E}$ can be represented as a column-stacked matrix $\mathbf{E}$ of size $d_{\text{CTC}}^2 \times d_{\text{CTC}}^2$, with eigenvalues $\{\lambda_n\}$. The largest, $|\lambda_1| = 1$, corresponds to the fixed-point (or cycle-fundamental) mode. The second-largest, $|\lambda_2|$, controls the asymptotic convergence rate.

The Kraus decomposition for pure $\sigma_{\text{CR}} = |0\rangle\langle 0|$ gives d_{CR} operators $\{K_a\}_{a=1}^{d_{\text{CR}}}$ acting on $\mathcal{H}_{\text{CTC}}$, with $\mathcal{E}(\rho) = \sum_a K_a \rho K_a^\dagger$.

2.2 State-distinguisher amplification

For two close mixed states σ_a, σ_b on $\mathcal{H}_{\text{CR}}$ with trace distance $\|\sigma_a - \sigma_b\|_1/2 = \epsilon$, the Helstrom [4] success probability for ordinary quantum measurement on the inputs is $\frac{1}{2} + \frac{\epsilon}{2}$.

The D-CTC *amplification factor* for a channel U is

$$A_U = \frac{\|\rho_a^* - \rho_b^*\|_1}{\|\sigma_a - \sigma_b\|_1}, \quad (3)$$

where ρ_a^*, ρ_b^* are the fixed points (or cycle averages) of $\mathcal{E}$ conditioned on σ_a and σ_b . The post-D-CTC success probability is $\frac{1}{2} + A_U \epsilon/2$, so $A_U > 1$ means D-CTC iteration enhances distinguishability beyond direct quantum measurement.

3 Spectral oracle for convergence

3.1 Pearson correlation $r = 0.99$

For 2000 Haar-random unitaries we measured the empirical iteration count N_ϵ to converge to tolerance ϵ and compared to the textbook mixing-time prediction

$$N_\epsilon^{\text{predicted}} = \frac{-\log \epsilon}{\log |\lambda_2(\mathbf{E})|^{-1}}. \quad (4)$$

Across the sample we find Pearson correlation $r = 0.99$, log-log slope 0.92 (theoretical: 1.0). The slight under-1 slope reflects sub-leading-eigenvalue corrections that accelerate the late-stage decay beyond the strict $|\lambda_2|$ asymptote.

3.2 Log-normal iteration distribution

The iteration count distribution itself is log-normal (Akaike Information Criterion advantage $\Delta\text{AIC} = 2101$ over a maximum-likelihood exponential fit). Log-normal arises naturally from multiplicative cascade dynamics: each iteration multiplies the residual by an effectively-stochastic factor, and the central limit theorem applied to the *logarithm* of iteration count gives the observed distribution shape. The tail follows an empirical power law with exponent $\alpha \approx 4$ (finite variance, heavy tail).

3.3 Power-law dimensional scaling

The median iteration count satisfies $\langle N_\epsilon \rangle \sim d_{\text{CR}}^{-0.85}$ for $d_{\text{CR}} \in \{2, \dots, 8\}$ and $d_{\text{CTC}} \in \{3, 4, 5\}$. The exponent is essentially flat across d_{CTC} , suggesting a universal scaling of the spectral gap with CR register size.

4 Aaronson–Watrous amplification: empirical

4.1 Channel-class enhancement

The fraction of Haar-random unitaries producing fixed-point purity > 0.9 is approximately 0.4% at $d_{\text{CR}} = 2$, $d_{\text{CTC}} = 3$. By contrast, the subgroup of unitaries of the form $U = P \cdot D$ where P is a permutation matrix and $D = \text{diag}(\pm 1, \pm i, \dots)$ produces purity > 0.9 in $\approx 40\%$ of samples — a $100\times$ rate enhancement.

The permutation-times-fourth-roots subgroup is a strict sub-group of the Clifford group, the natural algebraic class for stabiliser-state quantum information processing.

4.2 Amplification scales with input closeness

The empirical amplification grows sharply as the two input states approach (Table 1). For inputs at $\epsilon = 0.05$ where the Helstrom bound is barely above guessing (53%), optimised Clifford-like channels achieve $\sim 83\%$ success.

Table 1: Best speedup over Helstrom for Clifford-like vs Haar D-CTC channels. Speedup is defined as $(P_{\text{CTC}} - 0.5)/(P_{\text{Helstrom}} - 0.5)$.

ϵ	P_{Helstrom}	$P_{\text{CTC, Cliff.}}$	$P_{\text{CTC, Haar}}$	Cliff. speedup	Haar speedup
0.05	0.525	0.833	0.580	13.3	3.2
0.10	0.550	0.874	0.593	7.5	1.9
0.20	0.600	0.833	0.683	3.3	1.8
0.50	0.750	0.833	0.819	1.3	1.3

4.3 Noise robustness

At input $\epsilon = 0.1$ with 10% depolarising noise per iteration step, the Clifford D-CTC speedup falls from $1.82\times$ (noiseless) to $1.52\times$, a $\sim 17\%$ reduction. The amplification is therefore practically robust, not just an idealised zero-noise phenomenon.

5 Limit cycle structure

The standard fixed-point reading of (1) is incomplete: most Clifford-class channels do not converge.

5.1 Cycle-length distribution

Across 1000 unitaries per ensemble:

Table 2: Cycle-length distribution at $(d_{\text{CR}}, d_{\text{CTC}}) = (2, 3)$.

Cycle length	Haar	Clifford-like
1 (converged)	65.9%	32.7%
2	17.5%	39.4%
3	5.8%	14.4%
4	0.3%	10.3%
≥ 5 or undetected	10.5%	0.0%

The majority of Clifford-class D-CTC channels reach a limit cycle, not a fixed point.

5.2 Spectral prediction of cycle length

The cycle length k is predicted exactly by the second-largest channel-superoperator eigenvalue:

$$\boxed{\arg \lambda_2(\mathbf{E}) = \pm \frac{2\pi}{k}.} \quad (5)$$

Empirically: period-3 channels have $\langle |\arg \lambda_2| \rangle = 2.094$, the textbook value of $2\pi/3 \approx 2.094$ to four decimal places. Period-2 channels cluster near π .

Critically, period- k channels have $|\lambda_2| = 1.000$ exactly — the subdominant mode is *on* the unit circle, neither decaying nor growing. This marginal mode preserves input information indefinitely.

5.3 Cycles amplify more than fixed points

Comparing the two cycle classes on the standard $\epsilon = 0.1$ amplification benchmark:

- Period-1 (fixed-point) channels: mean amplification 0.21
- Period-2 (cycling) channels: mean amplification 0.43

Cycle solutions amplify $\sim 2\times$ more than fixed-point solutions. The marginal-mode information preservation is the structural mechanism enabling Aaronson–Watrous amplification.

6 Multi-basin dynamics

A single Clifford-class D-CTC channel typically admits multiple distinct limit cycles, with the initial state selecting which one is reached.

6.1 Quantitative overturning of ergodicity

Using 30 diverse initial conditions per channel (random pure states, random mixed states, computational-basis states, pairwise superpositions, maximally-mixed state):

A typical Clifford-class channel has approximately 25 distinct limits, depending on initial state. Within a single cycle the iteration is ergodic: from 20 random pure inits all reaching a period-2 channel, both phases of the cycle are visited equally ($A = 10$, $B = 10$). The non-ergodicity is across-cycle, not within-cycle.

Table 3: Number of distinct limit states per channel; $n = 200$ channels per ensemble.

Ensemble	Mean distinct limits	Strict-unique fraction
Haar	1.0	100%
Clifford-like	24.8	18%

6.2 Adversarial probing

For a single typical Clifford channel, seven canonical initial states (basis states, pairwise superpositions, maximally-mixed) yielded at least four distinct limits with purities $\{1.0, 1.0, 0.5, 0.56\}$. The same channel produces *qualitatively different outputs* for different inputs — not merely the same fixed-point map evaluated on different inputs.

7 Chronology-protection coupling

We test whether D-CTC amplification persists at configurations where the classical CTC content is suppressed — the regime where Hawking’s chronology-protection conjecture [3] predicts quantum suppression of physical CTCs.

7.1 Setup

Using the Systrophe-pair offset parameter [8], we sweep $\delta \in [0, 2\pi]$. The pair’s classical CTC content (integrated $L^2(r)$) reaches maximum at $\delta = 0$ and exactly zero at $\delta = \pi$ — the chronology-extinction phase. We construct $U(\delta)$ in three qualitatively different ways and measure the D-CTC amplification at each δ .

7.2 Three encoding regimes

Table 4: Encoding-regime taxonomy of D-CTC channel constructions.

Regime	Construction	Amp. at $\delta = \pi$
Abstract	Clifford, Floquet, generic Hamiltonian	varies; often peak
Direct-CTC	matrix entries $\propto L_{\text{pair}}(r)$	0 (vanishes)
LP-Hamiltonian	$H(F, K, L)$ on Clifford backbone	0.18 (constant)

Abstract encodings are decoupled from the underlying spacetime CTC content (mean Pearson -0.02 across four distinct constructions). **Direct-CTC encodings** inherit chronology protection by structural degeneracy: when $L_{\text{pair}}(\delta)$ vanishes, the matrix collapses and the channel becomes trivial. **LP-Hamiltonian encodings** achieve a perfect purity of 1.0 but only modest amplification.

7.3 The encoding is a continuum

Hybrid constructions

$$U(\delta, t) = \text{polar}[(1 - t)U_{\text{Clifford}} + t \cdot U_{\text{direct-L}}(\delta)] \quad (6)$$

show a smooth transition. The chronology-protection signature (amplification ratio ≤ 0.5 at $\delta = \pi$ vs $\delta = 0$) appears at $t \geq 0.5$ and grows monotonically to full signature at $t = 1$. There is no sharp phase boundary; the three regimes of Table 4 are limit points of a smooth continuum parametrised by the channel’s direct-CTC content.

7.4 Implication

The Aaronson–Watrous PSPACE speedup is *contingent on channel encoding*, not on physical CTC existence. Concretely:

- Implementations via post-selection circuits or pre-engineered Clifford gates: AW speedup persists at chronology-protected configurations.
- Implementations that derive the channel matrix from the spacetime CTC tensor literally: AW speedup is suppressed together with the CTCs.
- Physically grounded LP-Hamiltonian implementations: AW speedup is muted but constant across the regime.

8 Mechanism: the Clifford structural signature

Five structural features distinguish the Clifford channel class from generic (Haar) channels:

8.1 Bimodal eigenvalue distribution

The $|\lambda|$ distribution of Clifford channels is bimodal: 65% of eigenvalues in $[0, 0.1]$ and 35% in $[0.9, 1.0]$, with *nothing in between*. Haar channels have a smooth bulk distribution. The bimodality means subdominant modes decay rapidly (small $|\lambda|$) while the principal mode and (when present) the marginal mode are quasi-degenerate at the unit circle.

8.2 Structurally degenerate U-blocks

Treating U as a tensor on $\mathcal{H}_{\text{CR}}^{\text{out}} \otimes \mathcal{H}_{\text{CTC}}^{\text{out}} \otimes \mathcal{H}_{\text{CR}}^{\text{in}} \otimes \mathcal{H}_{\text{CTC}}^{\text{in}}$, each (b, a) -slice $U[b, :, a, :]$ is a $d_{\text{CTC}} \times d_{\text{CTC}}$ matrix. For Clifford-like $U = P \cdot D$, most of these slices are all-zero (the permutation matrix has a single 1 per row). The empirical mean condition number of non-zero slices is $\sim 10^{12}$ for Clifford vs. ~ 7 for Haar.

8.3 Full Bell-state disentanglement

For a Bell-like state on $(\mathcal{H}_{\text{CR}} \otimes \mathcal{H}_{\text{CTC}}^{(0)})$ where $\mathcal{H}_{\text{CTC}}^{(0)}$ denotes a two-dim subspace of $\mathcal{H}_{\text{CTC}}$, the post- U reduced CR entropy satisfies:

- Clifford-class U : full disentanglement ($\Delta S = -0.69$, the maximum possible)
- Haar-random U : partial disentanglement ($\langle \Delta S \rangle = -0.21$)

The Clifford full disentanglement is the structural mechanism behind state-distinguisher amplification: the channel deterministically extracts CR information into a definite $\mathcal{H}_{\text{CTC}}$ subspace.

8.4 Higher Holevo classical capacity

Estimated Holevo classical capacity (using three computational-basis inputs with uniform weights):

- Clifford-like: 0.83 (out of $\log 3 = 1.10$ theoretical max)
- Haar: 0.53

Clifford channels carry $\sim 60\%$ more classical information, consistent with their information-preserving marginal modes.

8.5 Code-state-quality fixed points

Approximately 33% of Clifford-class channels produce rank-1 (purity > 0.99) fixed points usable as quantum error-correction code states. This naturally couples D-CTC dynamics to stabiliser-state quantum information.

9 Re-interpretation of the Deutsch equation

The six empirical regularities suggest a revised reading of (1). We propose that Deutsch’s self-consistency equation should admit cycle solutions of any period $k \geq 1$:

$$\rho_n = \rho_{n+k}, \quad k = 1, 2, 3, 4, \dots, \quad (7)$$

with $k = 1$ being strict fixed points and $k \geq 2$ being limit cycles.

9.1 Properties of the cycle-solution class

Cycle solutions of period $k \geq 2$ exhibit:

- Generic existence in the Clifford structural class (combined incidence 63% at $d = 6$);
- Spectral signature $\arg \lambda_2(\mathbf{E}) = 2\pi/k$ with $|\lambda_2| = 1.000$ exactly;
- Marginal-mode quantum-information preservation;
- Mean amplification roughly $2\times$ the fixed-point class;
- Multi-basin selection: initial state determines which cycle is reached.

9.2 Aaronson–Watrous on the cycle class

The Aaronson–Watrous polynomial-time PSPACE result [2] applies naturally to cycle solutions: the algorithm only requires self-consistency, not strict convergence, and (1) admits both. The cycle class is in fact the *empirical* site of the amplification: period-2 channels amplify twice as much as period-1 channels.

9.3 Numerical practice

Standard D-CTC simulators use a fixed-point convergence test $\|\rho_{n+1} - \rho_n\| < \epsilon$. *This test reports false positives on cycle solutions of any period:* since within a period- k cycle consecutive iterates are within tolerance of each other if the trajectory traverses the cycle slowly. Explicit cycle detection (retaining the last $k = 1, 2, \dots$ iterates and testing pairwise distances) is necessary. The Systrophe package’s `detect_cycle_length` routine provides this.

10 Implications

10.1 Computational complexity

The Aaronson–Watrous result is intact but refined: the relevant channel class is the cycle-supporting Clifford-like subgroup, not generic Haar randomness. Implementing this via post-selection quantum circuits on NISQ hardware should be tractable and would test the AW prediction directly.

10.2 Physics interpretation

If Deutsch’s equation is read literally as a constraint on the quantum state “at the closed timelike curve”, cycle solutions say the state need not be unique — it can be a periodic trajectory through a finite-dim subspace of density matrices. The Hawking chronology-protection conjecture [3] acts on the classical-CTC layer, suppressing the spacetime CTC tensor; the D-CTC amplification mechanism lives at the channel-algebra level, which is contingently coupled to the spacetime layer through the channel-encoding choice.

10.3 Numerical practice

Always check for cycles explicitly. The standard fixed-point convergence test reports false positives. The Systrophe package provides cycle-detection routines and spectral-oracle convergence prediction as public APIs.

11 Open questions

1. Is the Clifford-class amplification a smooth function of “Clifford distance”? Our X -phase test gave $r \sim 0.08$ suggesting no, but the distance metric may need refinement.
2. Does the bimodal $|\lambda|$ distribution have a theoretical explanation in random-matrix theory? The peak at $|\lambda| \approx 0$ in particular may be related to the structurally-degenerate U -blocks of permutation-times-diagonal unitaries.
3. Can higher-period cycles ($k \geq 3$) be useful for PSPACE algorithm design? The amplification correlates with cycle period in our data, suggesting yes.
4. Is the smooth encoding continuum a topological invariant of the channel parametrisation, or a metric property of the unitary group near the Clifford submanifold?
5. Does the multi-basin structure imply that practical D-CTC implementations need a *protocol* for basin selection, e.g. post-selection on a specific cycle phase?

12 Reproduction

The 33 phase scripts are at github.com/Zynerji/systrophe/examples/dctc_deep_phase_*.py (v0.17.0). For the headline findings:

```
python examples/dctc_aw_amplification_demo.py
python examples/dctc_deep_phase_a.py           # scaling
python examples/dctc_deep_phase_b.py           # spectral oracle
python examples/dctc_deep_phase_d.py           # log-normal
python examples/dctc_deep_phase_i.py           # Clifford comparison
python examples/dctc_deep_phase_am_extended.py # encoding regime
python examples/dctc_deep_E1_cycles.py          # cycle dynamics
python examples/dctc_deep_E4_trichotomy.py      # encoding continuum
python examples/dctc_deep_E8_ergodicity.py      # multi-basin
```

Each writes JSON results to its corresponding `dctc_deep_*_results.json` file.

Acknowledgements

This work emerged from sustained empirical exploration with the Anthropic Claude coding assistant. The author conceived the overall investigation, ranked and ordered the experimental phases, and synthesised findings into theoretical claims. The assistant implemented and ran the simulations and contributed to interpretation. All numerical claims are reproducible from the published scripts.

References

- [1] D. Deutsch. Quantum mechanics near closed timelike curves. *Phys. Rev. D*, 44:3197–3217, 1991.
- [2] S. Aaronson and J. Watrous. Closed timelike curves make quantum and classical computing equivalent. *Proc. Roy. Soc. A*, 465:631–647, 2009.
- [3] S. W. Hawking. Chronology protection conjecture. *Phys. Rev. D*, 46:603–611, 1992.
- [4] C. W. Helstrom. *Quantum Detection and Estimation Theory*. Academic Press, 1976.
- [5] S. Aaronson. Quantum computing, postselection, and probabilistic polynomial-time. *Proc. Roy. Soc. A*, 461:3473–3482, 2005.
- [6] C. Knopp. Clifford-structured Deutsch-CTC channels: empirical amplification, spectral oracle, and the encoding-dependence of chronology protection. https://github.com/Zynerji/systrophe/blob/main/paper/dctc_amplification.pdf, 2026.
- [7] C. Knopp. Deutsch-CTC is a limit-cycle theory: numerical evidence. https://github.com/Zynerji/systrophe/blob/main/paper/dctc_cycles.pdf, 2026.
- [8] C. Knopp. Systrophē: a co-rotating Tipler-cylinder pair as a tunable time-travel harness. <https://github.com/Zynerji/systrophe>, 2026.